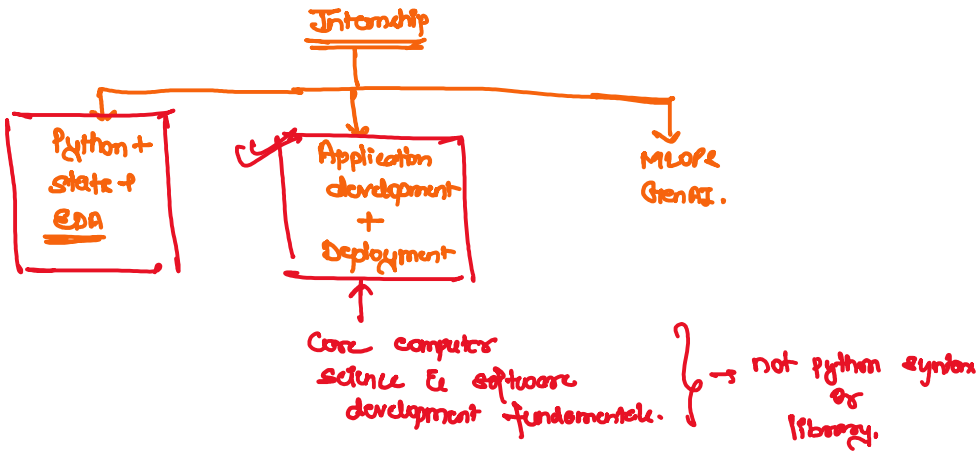
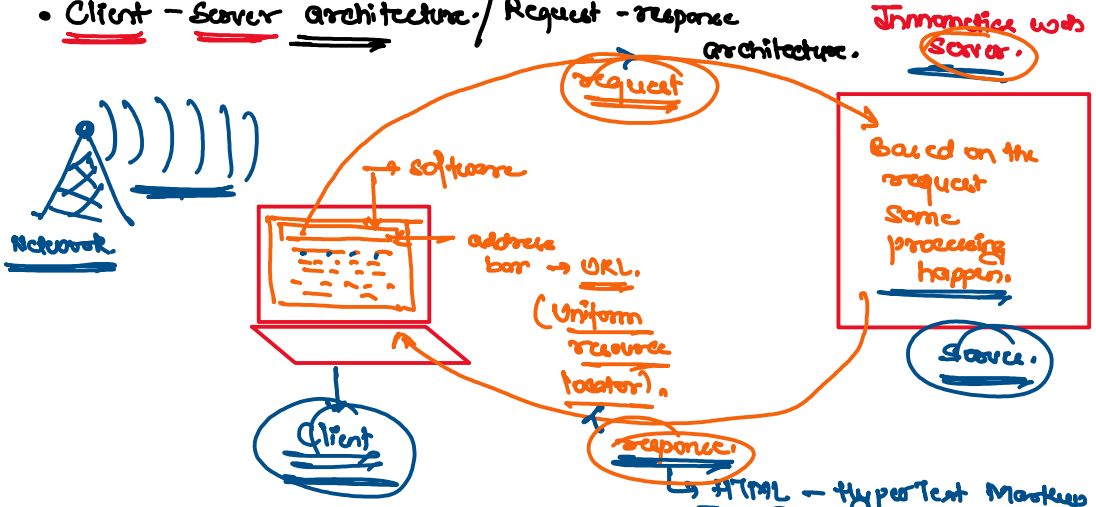
Friday, 26 December 2025 6:01 PM



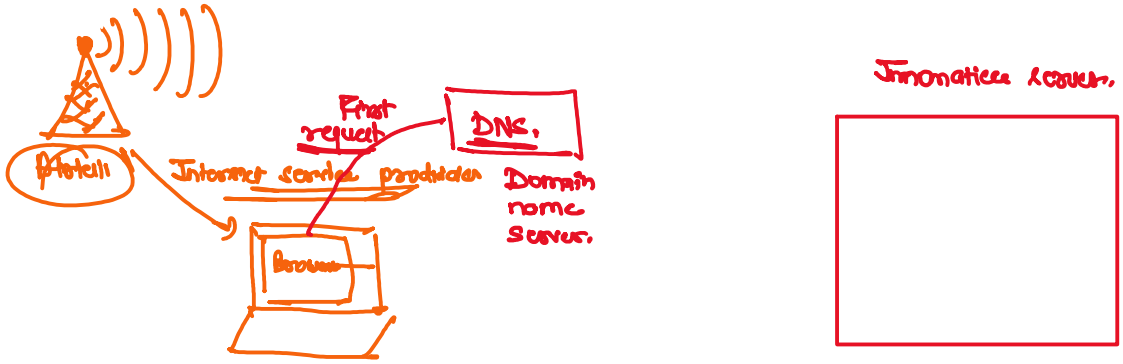
- Client - Server architecture / Request - response architecture.

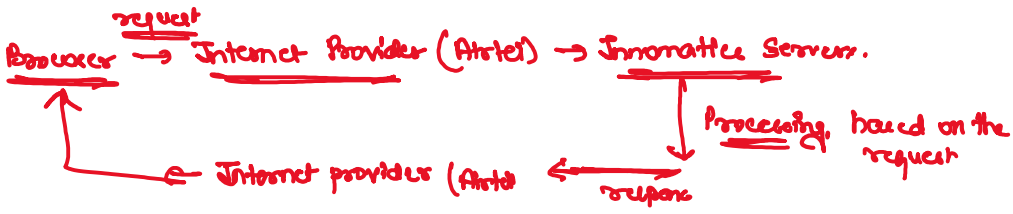


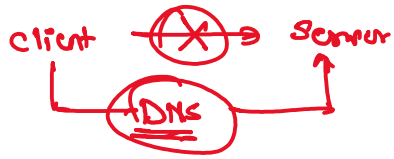
Processing could include
- Retrieving data.
- Running some backend logic.
- Preparing the webpage content

- A client can be
  - a laptop
  - a desktop.
  - a mobile phone
  - a tablet.

client - anything that can send a request & receive a response is a client.

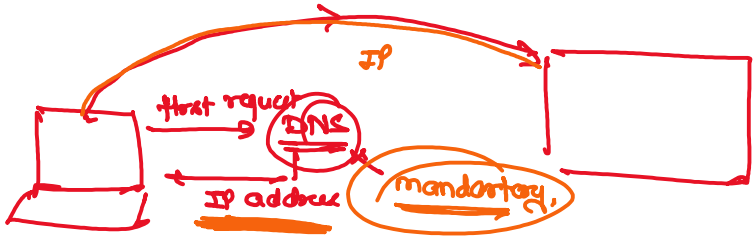




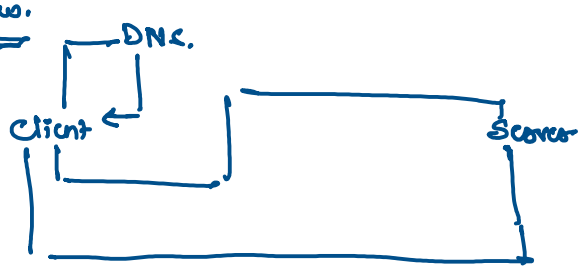


- DNS → Domain Name Server.
  - Address of a server → IP address
    - Internet protocol.
    - IP address is a unique address assigned to every server on the internet
- DNS stores a mapping.

  Domain Name → IP address.



- Actual flow.



Client → DNS → Server → Response → Client.

- Can Airtel or any other service provider see the data that we send in the request.?, → NO

  website → Facebook or Innomatics.

- Request contains
  - Your email ID or mobile number
  - Your password.

HTTP

HTTPS

HTTPS → secure.
→ data is encrypted.

HTTP

- If the server IP is stored in DNS and DNS belongs to the internet provider, what happens if we use a different internet provider.

Every internet provider must either.
- maintain their own DNS server
Or use third party DNS server.

Requests are send using.
- GET
- POST.

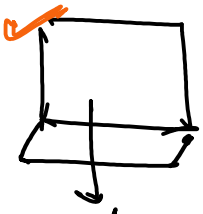
DNS → itself is a server.
Eg. Google public DNS : 8.8.8.8.

The response of DNS → it is always be an IP address

- Server
  → it is not just a database.

→ A server is a machine., it is a computer, just like a laptop, but much more powerful.

- A server is designed to
  - handle multiple requests at the same time.
  - Serve many user simultaneously.



- CPU / GPU
- Memory (RAM)
- Storage. → Secondary memory
- Network interface.

Server
- CPU | GPU
- memory.
- storage
  - Network interface
  - Special memory layer
- Cache.

Server are optimized. to handle scale, speed & multiple user